



Unity University

College of Engineering, Technology and Computation Sciences

Department of Computer Science

ParkIT

An automated parking system with client application

Chapter Two – Project Management

Group Members:

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Chapter II

Introduction – Project Management

Project Management Institute (PMI), has defined project management as “the application of knowledge, skills tools and techniques to a broad range of activities in order to meet the requirements of a particular project.”

Project management primarily focuses on planning, managing and organizing the available resources. Some of the activities that should be part of project management activity are to efficiently guide the project team through all phases and execute the project successfully. Other activities include identifying and implementing it to the user-centered design process.

Goals and Objectives Project Management

The objectives and goals of the organization can be achieved in an efficient manner through proper project management. Project manager plays a key role in managing the project. They forecast all the risks that might arise during the production process and ensure that all the risks are addressed with a proper plan. By following a good project management structure the employees will have a good understanding about their roles and responsibilities, they will also be aware of the schedule of the deliverables and will have the opportunity to plan their schedule accordingly. A good project management structure ensures that the project is completed within specified time period and within budget.

Work Breakdown Structure

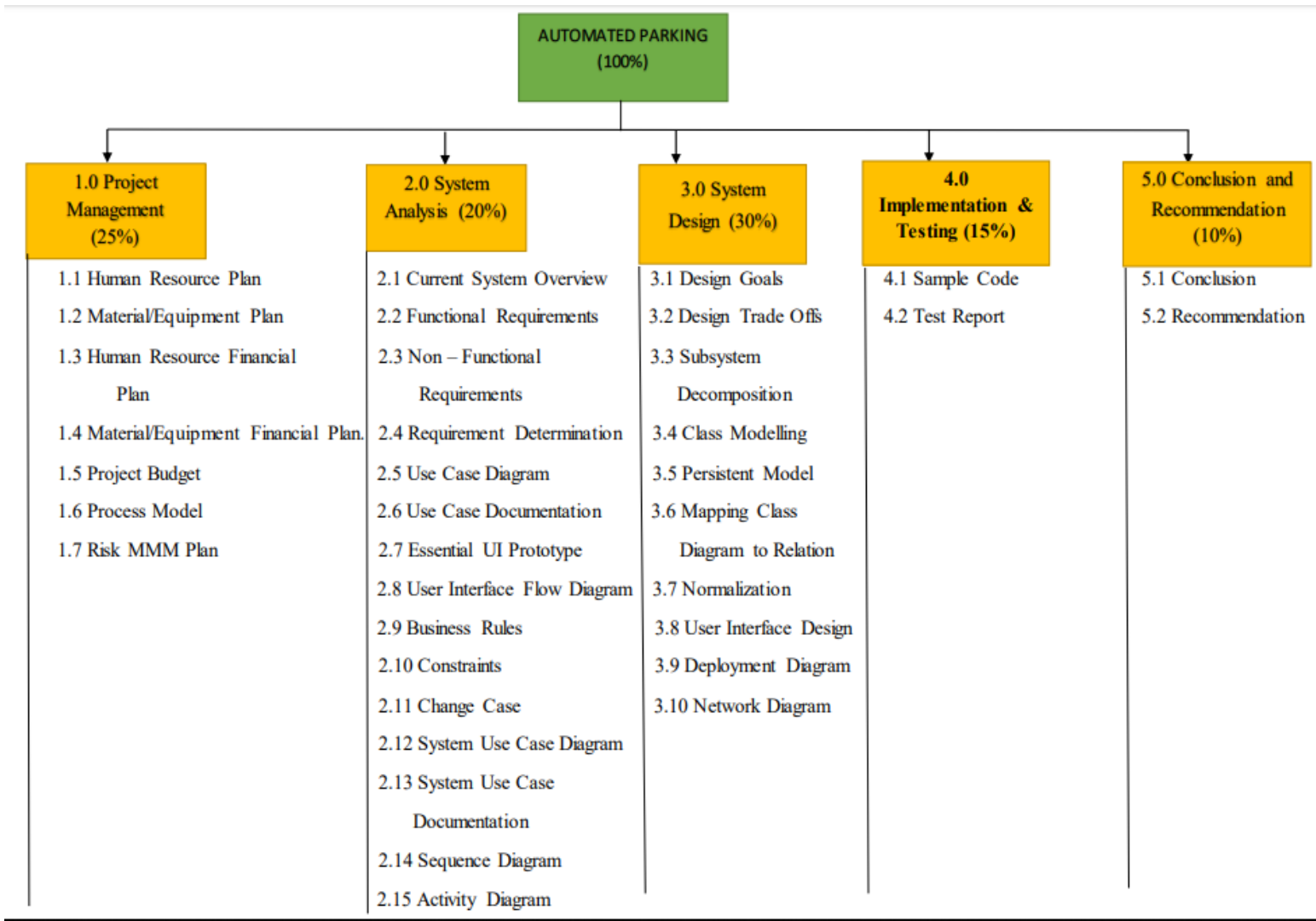


Fig 2.0 WBS

Human Resource Plan

<u>Job Title</u>	<u>Qualification</u>	<u>Activity List</u>	<u>Position Description</u>	<u>Names</u>	<u>Total</u>
Project analyst	BSC	2.0	The project analyst role leads and coordinates requirements elicitation and use-case modeling by outlining the system's functionality and delimiting the system.	Kirubel Getachew Natnael Behailu	2
Project manager	BSC	All Activities	The project manager role allocates resources, shapes priorities, coordinates interactions with customers and users, and generally keeps the project team focused on the right goal. The project manager also establishes a set of practices that ensure the integrity and quality of the project	Robel Gedamu	1
Software programmer	BSC	4.1	The Software programmer role is responsible for developing and testing components, in accordance with the projects adopted standards, for integration into larger subsystems. When test components, such as drivers or stubs, must be created to support testing, the Software programmer is also responsible for developing and testing the test components and corresponding subsystems.	Leul Dantew Bereket Awegchew Kirubel Getachew	3
System Designer	BSC	3.0	The System designer's role is to create the design that will be used for the system and the user interface design.	Bereket Awegchew Leul Dantew	2
Tester	Not Required	4.2	The Tester role is responsible for the core activities of the test effort, which involves conducting the necessary tests and logging the outcomes of that testing. This covers:		--

Material/Equipment Plan

No	Item Name	Items Description
Hardware		
1.	Computers	1. Toshiba c55b Ram-8gb HDD-500gb 2. Hp Pavilion G6 Ram-8gb HDD-700gb 3. Lenovo Notebook Ram-8gb HDD-1Tb 4. Dell G5 Ram-16 SSD-250gb HDD-2Tb
2.	Storage Devices	1. HDD 1Tb 2. Flash Drive 32gb
3.	Camera	1. Cannon EOS 90D 2. Webcam Camera 16MP
Software Development Tools		
1.	Text Editor	1. Visual Studio Code 2. Sublime Text Editor 3. Microsoft Office
2.	Graphics Designer	1. Adobe Illustrator 2. Adobe Photoshop 3. Adobe XD
3.	UI software development kit	1. Flutter 2. React
4.	Project Management tool	1. GitHub
Server		
1.	Database Server	1. Firebase

Financial Plan

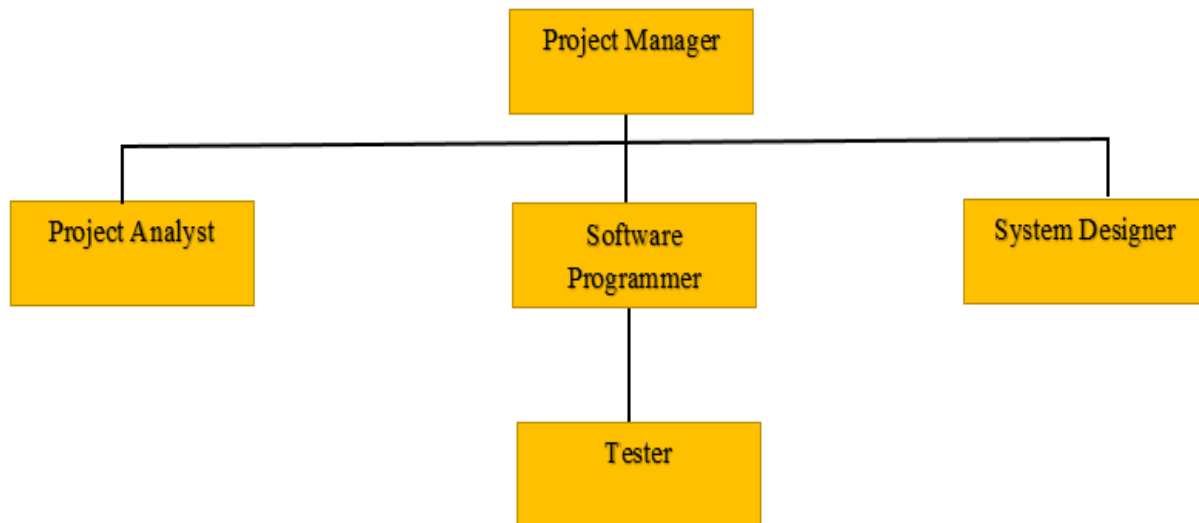
No	Roles	Mass	Total (in Birr)
Material/Equipment			
1.	Laptop	4	60,000
2.	Servers	1	Free
3.	Software resources	8	Free
4.	Storage Devices	2	3,150
5.	Camera	2	60,000
		Total	123,150

Project Budget

Our project budget not including the transportation and team meetings fee is currently estimated at 123,150 Birr.

Team Organization

We use a hierarchical Structured team organization.



Process Model

We use the Waterfall Model for our project.

The **Waterfall model** is the earliest SDLC approach that was used for software development. The **waterfall Model** illustrates the software development **process** in a linear sequential flow. This means that any phase in the development process begins only if the previous phase is complete.

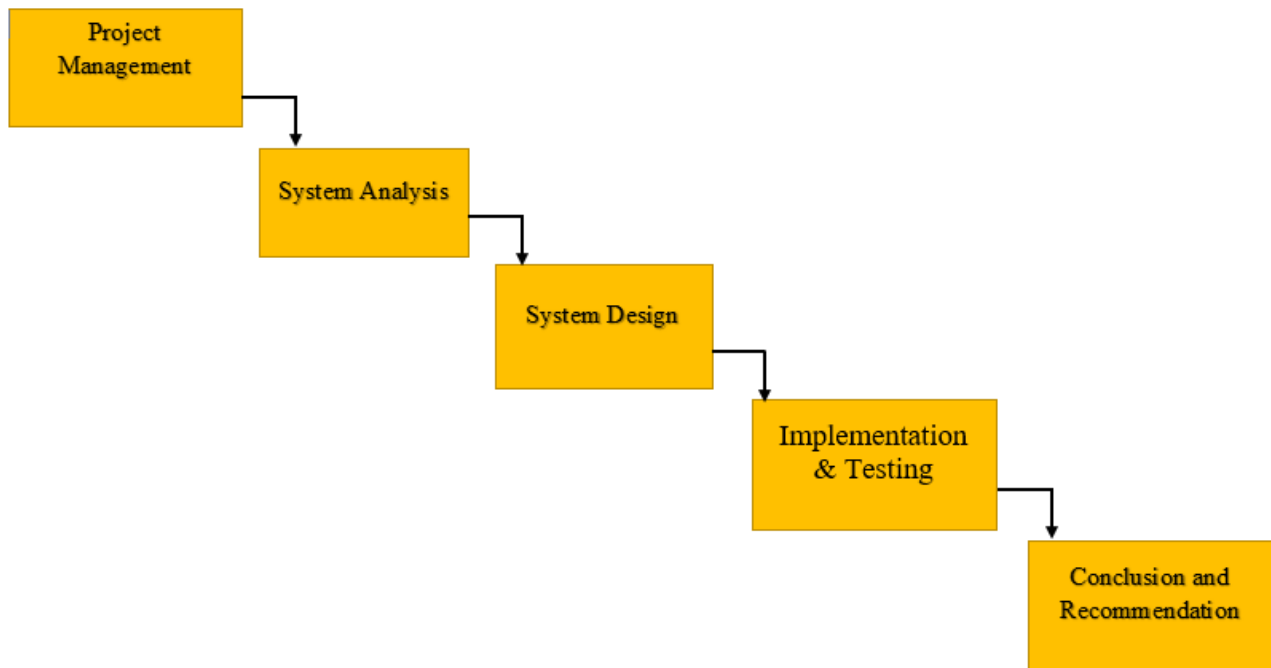


Fig 1.6 Waterfall Process Model

Risk MMM Plan

Risk Item Table

Risk Id No	Name of Risk	Risk category	Probability	Rank	Cause of risk
R1	Team Member Unavailability	People risk	15%	High	Some members of the team will be unavailable for certain project activities due to another work, illness or emergency
R2	Mismanagement	Structure or process risk	5%	Low	Due to lack of experience and may lead to miscommunication
R3	Changes to Project Scope	Structure or Process risk	15%	High	Team member may add or reduce project scope
R4	Unrealistic schedules	People risk	15%	High	A delay in one task may cause cascade delays in dependent tasks.
R5	Wrong cost estimation	Financial risk	15%	High	Due to lack of experience
R6	Sponsors late	Financial risk	15%	High	Lack of budget

Risk Mitigation

Risk Id No	Name of Risk	Strategies to Mitigate
R2	Mismanagement	Provide training
R3	Changes to Project scope	Evaluate the new changes and either add or remove scopes

Risk Monitoring

Risk Id No	Name of Risk	Strategies to monitor
R1	Team Member Unavailability	Reach out to current member or recruit new member

Risk Management

Risk Id No	Name of Risk	Strategies to manage risk
R4	Unrealistic schedules	Provide overtime
R5	Wrong cost estimation	Limit expense or ask for more budget
R6	Sponsors late	Pause project activities